

Technical Stack

Languages & Frameworks: Python (Jupyter, Numpy, Pandas, Dash, SciPy), Clojure, Bash, R, Java (Micronaut, Spring)

Data & Infrastructure: PostgreSQL (window functions, UDFs, PL/pgSQL, recursive CTEs), BigQuery, GCP (Google Logging, Pub/Sub, GCS), Kafka, Splunk, Spark SQL, Docker, architecture (Kimball/star, snowflake, xNF)

Visualization Frameworks: Looker, Google Logging, Splunk, Dash, Plotly, R-Shiny, Seaborn, Matplotlib

AI/ML: Vertex AI (Gemini), OpenAI, Cursor (Claude Sonnet/Opus, Gemini Code Assist), Scikit-learn, TensorFlow

Professional Experience

Rally Software

Data Engineer, Senior Data Engineer

CA Agile Central

Data Scientist, Senior Data Scientist, Principal Data Scientist

Broadcom (Rally – Agile Operations Division)

Staff Data Scientist

Broomfield, Denver, Boulder, CO

Oct 2015 – Oct 2025

- Led technical side of development for Agile flow metrics and Agile forecasting products.
- Developed a "Smart" Cumulative Flow Diagram for analyzing team and train delivery metrics.
- Led team and train-level portfolio flow metrics design and development.
- Used Vertex AI (Gemini) to produce flow metrics chart interpretations and coaching advice for teams.
- Spearheaded capacity and delivery forecasting development for team and train-level planning.
- Helped develop, maintain, and use a distributed tracing infrastructure for a visible and reliable usage and performance metrics system to report 100% of user interactions in a SaaS environment involving 250K+ monthly users. Used the resulting data with Splunk, BigQuery, Looker, Spark, and other systems to analyze interactions.
- Led performance optimization, via tracing and visualizations, for Rally's Oracle to PostgreSQL migration.
- Built and maintained a "product data hub" for those within Rally to track feature adoption, subscription usage, performance, and user input.
- Built a leading indicator for subscription renewal predictions using usage data and random forest models.
- Worked in Python, Java, Clojure, Bash, Scala, JavaScript (React), and elsewhere as needed.

SpotXChange

Data Scientist

Broomfield, CO

Apr 2014 – Sep 2015

- Developed a large-scale Hive datastore for 600+ frontend ad-tech bidding servers using Python and Bash.
- Tracked automated view bidding for website advertisements.
- Led development of fraud activity investigations using big data, Hadoop, ETL, MySQL, and Tableau.
- Built Tableau dashboards for product and C-level insights.

Silicon Graphics, Inc. (SGI)

Software Engineer

Longmont, CO

Jun 2013 – Mar 2014

- Led development and testing of SCSI Enclosure Service (SES) Python/Cython module.
- Worked on various firmware/control modules in C, C++ (Boost), Python, and Cython.

Education

Ph.D. in Mathematics (2014/ABD)

University of Colorado (Alexander Hulpke and Keith Kearnes)

- Computational group and graph theory, permutation group algorithms, NCAR & XSEDE supercomputers
- Developed a massively parallel approach to solving partition backtrack problems in group and graph theory.

M.A. in Mathematics – 2010

University of Colorado (Keith Kearnes)

- Computational group theory, representation theory

M.S. in Mathematics – 2006

University of Vermont (David S. Dummit)

- Computational algebraic number theory
- Developed an algorithm for finding number fields satisfying given properties and having minimum discriminant. Found the first known instance of a totally real quintic number field having minimal signature group rank.

B.S. in Mathematics – 2004 (logic, minor in physics)

University of Michigan–Flint

- Studied abroad in Uppsala, Sweden for one year; completed coursework in mathematics in Swedish and philosophy in English.
- President of student groups: Mathematics Club, Students for Social Change

Miscellaneous

Volunteering

- Volunteered at Elephant Nature Park in Thailand (2019, 2020, 2024, 2025) – assisted in the rescue, rehabilitation, and care of abused elephants.
- Volunteer with A Precious Child, Broomfield, CO (May 2026 – present) – donation center sorting/organizing.

Meetups and Knowledge Sharing

- I attend and have spoken four times at local Agile Meetup groups (Agile Boulder, Agile Denver, Agile Denver North), discussing: Cumulative Flow Diagrams (CFDs), probability and "wisdom-of-the-crowds" versus "group-think," Monte Carlo simulations for Agile forecasting, and some details about Little's Law.
- I coordinated the Denver/Boulder Type-1 Diabetic Adults Meetup from early 2017 until early 2020.
- I helped coordinate and spoke three times at the Analyze Boulder Meetup.
- Splunk invited me to give a talk in 2018. I presented on distributed tracing, ingestion into Splunk, and the resulting data analysis and visualizations that can be carried out on the platform.
- I coordinated the Denver Area Twitter Developer Community Meetup from October of 2016 until early 2020. I was not employed by Twitter, but we worked together to spread info about their APIs and analysis of Twitter data.

Other Employment

- In addition to teaching many courses in mathematics while attending graduate school, I held two adjunct instructor positions, one at Saint Michael's College in mathematics and one at The Community College of Vermont in photography.
- I worked multiple times as a contract photographer for Automobile Magazine.